

Dennis Sylvester

professor

基本信息

姓名: Dennis Sylvester

所属院校: 密歇根大学-安娜堡

地区: United States

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

数字集成电路设计

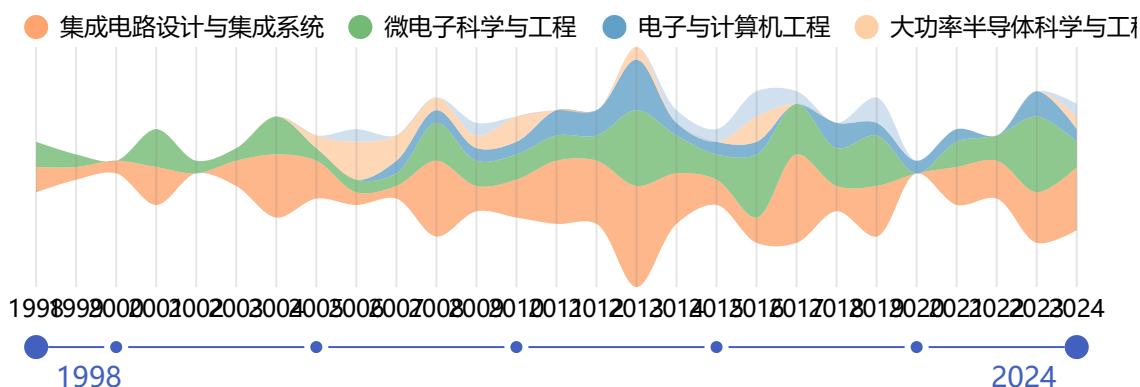
个人简介

Dennis Sylvester is a Professor of Electrical Engineering and Computer Science at the University of Michigan. His research focuses on integrated circuits and VLSI, with significant contributions to near-threshold computing, dynamic voltage scaling, and ultralow-voltage CMOS technology. He has authored numerous influential articles and is recognized for pioneering work in energy-efficient integrated circuits and low-power electronics. His email is verified at umich.edu.

工作经历

University of Michigan - Professor

近年发表的论文



AIMMI: Audio and Image Multi-Modal Intelligence via a Low-Power SoC With 2-MByte On-Chip MRAM for IoT Devices

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fan, Zichen; Zhang, Qirui; An, Hyochan; Xu, Boxun; Xu, Li; Tseng, Chien-Wei; Peng, Yimai; Bejarano-Carbo, Andrea; Abillama, Pierre; Cao, Ang; Liu, Bowen; Lee, Changwoo; Wang, Zhehong; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

被引用次数: 0

2024

RoboVisio: A Micro-Robot Vision Domain-Specific SoC for Autonomous Navigation Enabling Fully-on-Chip Intelligence via 2-MB eMRAM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Qirui; Fan, Zichen; An, Hyochan; Wang, Zhehong; Li, Ziyun; Wang, Guanru; Abillama, Pierre; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

被引用次数: 0

2024

A Sub-mm³ Wireless Neural Stimulator IC for Visual Cortical Prosthesis With Optical Power Harvesting and 7.5-kb/s Data Telemetry

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Jungho; Letner, Joseph G.; Lim, Jongyup; Atzeni, Gabriele; Liao, Jiawei; Kamboj, Abhilasha; Mani, Bhavika; Jeong, Seokhyeon; Kim, Yejoong; Sun, Yi; Koo, Beomseo; Richie, Julianna; Valle, Elena della; Patel, Paras R.; Sylvester, Dennis; Kim, Hun-Seok; Jang, Taekwang; Phillips, Jamie D.; Chestek, Cynthia A.; Weiland, James; Blaauw, David

被引用次数: 0

2024

An Ultralow-Power Triaxial MEMS Accelerometer With High-Voltage Biasing and Electrostatic Mismatch Compensation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Peng, Yimai; Jeong, Seokhyeon; Choo, Kyojin; Kim, Yejoong; Chen, Li-Yu; Rothe, Rohit; Xu, Li; Gurin, Ilya; Oliaei, Omid; Thompson, Matthew J.; Bart, Stephen; Hartwell, Peter; Blaauw, David; Sylvester, Dennis

被引用次数: 0

2024

A 1.5- μ W Fully-Integrated Keyword Spotting SoC in 28-nm CMOS With Skip-RNN and Fast-Settling Analog Frontend for Adaptive Frame Skipping

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Heejin; Seol, Ji-Hwan; Rothe, Rohit; Fan, Zichen; Zhang, Qirui; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

被引用次数: 2

2024

A 22nm 9.51 TOPS/W Neural Engine with 2MB MRAM Leveraging Sparse-Orthogonal Walsh-Hadamard Transform Computations and Dynamic Power Gating

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Abillama, Pierre; Zhang, Qirui; Fan, Zichen; An, Hyochan; Chen, Kuan-Yu; Chen, Yu; Blaauw, David; Sylvester, Dennis; Kim, Hun-Seok

被引用次数: 0

2024

A Fully Integrated, Automatically Generated DC-DC Converter Maintaining >75% Efficiency From 398 K Down to 23 K Across Wide Load Ranges in 12-nm FinFET

IEEE SOLID-STATE CIRCUITS LETTERS

Li, Anhang; Lee, Jeongsup; Mukim, Prashansa; Hoskins, Brian D.; Shrestha, Pragya; Wentzloff, David; Blaauw, David; Sylvester, Dennis; Saligane, Mehdi

被引用次数: 0

2024

An Ultralow-Power H.264/AVC Intra-Frame Image Compression Accelerator for Intelligent Event-Driven IoT Imaging Systems

IEEE SOLID-STATE CIRCUITS LETTERS

Zhang, Qirui; An, Hyochan; Bejarano-Carbo, Andrea; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

被引用次数: 0

2024

Quantum Circuit Simulation with Fast Tensor Decision Diagram

2024 25TH INTERNATIONAL SYMPOSIUM ON QUALITY ELECTRONIC DESIGN, ISQED 2024

Zhang, Qirui; Saligane, Mehdi; Kim, Hun-Seok; Blaauw, David; Tzimpragos, Georgios; Sylvester, Dennis

被引用次数: 0

2024

New Associate Editor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sylvester, Dennis

被引用次数: 0

2023

New Associate Editor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sylvester, Dennis

被引用次数: 0

2023

New Associate Editor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sylvester, Dennis

被引用次数: 0

2023

New Associate Editor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sylvester, Dennis

被引用次数: 0

2023

A High-Voltage Generator and Multiplexer for Electrostatic Actuation in Programmable Matter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Peng, Yimai; Carichner, Gordy; Kim, Yejoong; Chen, Li-Yu; Tribhout, Remy; Piranda, Benoit; Bourgeois, Julien; Blaauw, David; Sylvester, Dennis

被引用次数: 5

2023

TaskFusion: An Efficient Transfer Learning Architecture with Dual Delta Sparsity for Multi-Task Natural Language Processing

PROCEEDINGS OF THE 2023 THE 50TH ANNUAL INTERNATIONAL SYMPOSIUM ON COMPUTER ARCHITECTURE, ISCA 2023

Fan, Zichen; Zhang, Qirui; Abillama, Pierre; Shoouri, Sara; Lee, Changwoo; Blaauw, David; Kim, Hun-Seok; Sylvester, Dennis

被引用次数: 2

2023

SONA: An Accelerator for Transform-Domain Neural Networks with Sparse-Orthogonal Weights

2023 IEEE 34TH INTERNATIONAL CONFERENCE ON APPLICATION-SPECIFIC SYSTEMS, ARCHITECTURES AND PROCESSORS, ASAP

Abillama, Pierre; Fan, Zichen; Chen, Yu; An, Hyochan; Zhang, Qirui; Choi, Seungkyu; Blaauw, David; Sylvester, Dennis; Kim, Hun-Seok

被引用次数: 0

2023

A 4.6nW Subthreshold Voltage Reference with 400x Current Variation Reduction and 64-Step 0.11% Output Voltage Programmability

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Peng, Yimai; Bhat, Ashwin; Wadhwa, Sanjay; Blaauw, David; Sylvester, Dennis

被引用次数: 0

2023

Message From the Incoming Editor-in-Chief

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sylvester, Dennis

被引用次数: 0

2022

A Light-Tolerant Wireless Neural Recording IC for Motor Prediction With Near-Infrared-Based Power and Data Telemetry

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lim, Jongyup; Lee, Jungho; Moon, Eunseong; Barrow, Michael; Atzeni, Gabriele; Letner, Joseph G.; Costello, Joseph T.; Nason, Samuel R.; Patel, Paras R.; Sun, Yi; Patil, Parag G.; Kim, Hun-Seok; Chestek, Cynthia A.; Phillips, Jamie; Blaauw, David; Sylvester, Dennis; Jang, Taekwang

被引用次数: 13

2022

A 43 nW, 32 kHz, ± 4.2 ppm Piecewise Linear Temperature-Compensated Crystal Oscillator With $\Delta\sigma$ -Modulated Load Capacitance

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Sujin; Seol, Ji-Hwan; Xu, Li; Cho, Seonghwan; Sylvester, Dennis; Blaauw, David

A Delta Sigma-Modulated Sample and Average Common-Mode Feedback Technique for Capacitively Coupled Amplifiers in a 192-nW Acoustic Analog Front-End

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Rothe, Rohit; Cho, Minchang; Choo, Kyojin; Jeong, Seokhyeon; Oh, Sechang; Lee, Jungho; Sylvester, Dennis; Blaauw, David

被引用次数: 5

2022

A 510-pW 32-kHz Crystal Oscillator With High Energy-to-Noise-Ratio Pulse Injection

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xu, Li; Jang, Taekwang; Lim, Jongyup; Choo, Kyojin David; Blaauw, David; Sylvester, Dennis

被引用次数: 9

2022

A Power-Efficient Brain-Machine Interface System With a Sub-mw Feature Extraction and Decoding ASIC Demonstrated in Nonhuman Primates

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

An, Hyochan; Nason-Tomaszewski, Samuel R.; Lim, Jongyup; Kwon, Kyumin; Willsey, Matthew S.; Patil, Parag G.; Kim, Hun-Seok; Sylvester, Dennis; Chestek, Cynthia A.; Blaauw, David

被引用次数: 12

2022

An Open-Source and Autonomous Temperature Sensor Generator Verified With 64 Instances in SkyWater 130 nm for Comprehensive Design Space Exploration

IEEE SOLID-STATE CIRCUITS LETTERS

Zhang, Qirui; Duan, Wenbo; Edwards, Tim; Ansell, Tim; Blaauw, David; Sylvester, Dennis; Saligane, Mehdi

被引用次数: 15

2022

Reference Oversampling PLL Achieving -256-dB FoM and -78-dBc Reference Spur

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Seol, Ji-Hwan; Choo, Kyojin; Blaauw, David; Sylvester, Dennis; Jang, Taekwang

被引用次数: 16

2021

An Ultra-Low-Power Image Signal Processor for Hierarchical Image Recognition With Deep Neural Networks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

An, Hyochan; Schiferl, Sam; Venkatesan, Siddharth; Wesley, Tim; Zhang, Qirui; Wang, Jingcheng; Choo, Kyojin D.; Liu, Shiyu; Liu, Bowen; Li, Ziyun; Gong, Luyao; Zhong, Hengfei; Blaauw, David; Dreslinski, Ronald; Kim, Hun Seok; Sylvester, Dennis

被引用次数: 11

2021

RRAM-DNN: An RRAM and Model-Compression Empowered All-Weights-On-Chip DNN Accelerator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Ziyun; Wang, Zhehong; Xu, Li; Dong, Qing; Liu, Bowen; Su, Chin-I; Chu, Wen-Ting; Tsou, George; Chih, Yu-Der; Chang, Tsung-Yung Jonathan; Sylvester, Dennis; Kim, Hun-Seok; Blaauw, David

被引用次数: 24

2021

Fully-Autonomous SoC Synthesis Using Customizable Cell-Based Analog and Mixed-Signal Circuits Generation

VLSI-SOC: DESIGN TRENDS, VLSI-SOC 2020

Ajayi, Tutu; Kamineni, Sumanth; Fayazi, Morteza; Cherivirala, Yaswanth K.; Kwon, Kyumin; Gupta, Shourya; Duan, Wenbo; Lee, Jeongsup; Chen, Chien-Hen; Saligane, Mehdi; Sylvester, Dennis; Blaauw, David; Dreslinski, Ronald, Jr.; Calhoun, Benton; Wentzloff, David D.

被引用次数: 0

2021

A 28-nm Compute SRAM With Bit-Serial Logic/Arithmetic Operations for Programmable In-Memory Vector Computing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Jingcheng; Wang, Xiaowei; Eckert, Charles; Subramaniyan, Arun; Das, Reetuparna; Blaauw, David; Sylvester, Dennis

被引用次数: 135

2020

A Self-Tuning IoT Processor Using Leakage-Ratio Measurement for Energy-Optimal Operation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Jeongsup; Zhang, Yiqun; Dong, Qing; Lim, Wootae; Saligane, Mehdi; Kim, Yejoong; Jeong, Seokhyeon; Lim, Jongyup; Yasuda, Makoto; Miyoshi, Satoru; Kawaminami, Masaru; Blaauw, David; Sylvester, Dennis

被引用次数: 39

2020

Neural Cache: Bit-Serial In-Cache Acceleration of Deep Neural Networks

IEEE MICRO

Eckert, Charles; Wang, Xiaowei; Wang, Jingcheng; Subramaniyan, Arun; Iyer, Ravi; Sylvester, Dennis; Blaauw, David; Das, Reetuparna

被引用次数: 10

2019

An Efficient Piezoelectric Energy Harvesting Interface Circuit Using a Sense-and-Set Rectifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Peng, Yimai; Choo, Kyojin D.; Oh, Sechang; Lee, Inhee; Jang, Taekwang; Kim, Yejoong; Lim, Jongyup; Blaauw, David; Sylvester, Dennis

被引用次数: 35

2019

Energy-Efficient Motion-Triggered IoT CMOS Image Sensor With Capacitor Array-Assisted Charge-Injection SAR ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choo, Kyojin David; Xu, Li; Kim, Yejoong; Seol, Ji-Hwan; Wu, Xiao; Sylvester, Dennis; Blaauw, David

被引用次数: 34

2019

An Acoustic Signal Processing Chip With 142-nW Voice Activity Detection Using Mixer-Based Sequential Frequency Scanning and Neural Network Classification

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Sechang; Cho, Minchang; Shi, Zhan; Lim, Jongyup; Kim, Yejoong; Jeong, Seokhyeon; Chen, Yu; Rothe, Rohit; Blaauw, David; Kim, Hun-Seok; Sylvester, Dennis

被引用次数: 41

2019

A 1920 x 1080 25-Frames/s 2.4-TOPS/W Low-Power 6-D Vision Processor for Unified Optical Flow and Stereo Depth With Semi-Global Matching

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Ziyun; Wang, Jingcheng; Sylvester, Dennis; Blaauw, David; Kim, Hun Seok

被引用次数: 23

2019

A 42 nJ/Conversion On-Demand State-of-Charge Indicator for Miniature IoT Li-Ion Batteries

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Junwon; Jeong, Seokhyeon; Sylvester, Dennis; Blaauw, David; Kim, Chulwoo

被引用次数: 4

2019

A 1-Mb 28-nm 1T1MTJ STT-MRAM With Single-Cap Offset-Cancelled Sense Amplifier and In Situ Self-Write-Termination

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Dong, Qing; Wang, Zhehong; Lim, Jongyup; Zhang, Yiqun; Sinangil, Mahmut E.; Shih, Yi-Chun; Chih, Yu-Der; Chang, Jonathan; Blaauw, David; Sylvester, Dennis

被引用次数: 38

2019

An Area-Efficient 128-Channel Spike Sorting Processor for Real-Time Neural Recording With 0.175 μ W/Channel in 65-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Anh Tuan Do; Zeinolabedin, Seyed Mohammad Ali; Jeon, Dongsuk; Sylvester, Dennis; Kim, Tony Tae-Hyoung

被引用次数: 33

2019

A 256 pixel, 21.6 μ W infrared gesture recognition processor for smart devices

MICROELECTRONICS JOURNAL

Le Ba, Ngoc; Oh, Sechang; Sylvester, Dennis; Kim, Tony Tae-Hyoung

被引用次数: 5

2019

The Internet of Tiny Things: Recent Advances of Millimeter-Scale Computing

IEEE DESIGN & TEST

Oh, Sechang; Blaauw, David; Sylvester, Dennis

被引用次数: 14

2019

A Fixed-Point Neural Network Architecture for Speech Applications on Resource Constrained Hardware

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Shah, Mohit; Arunachalam, Sairam; Wang, Jingcheng; Blaauw, David; Sylvester, Dennis; Kim, Hun-Seok; Seo, Jae-sun; Chakrabarti, Chaitali

被引用次数: 24

2018

Recryptor: A Reconfigurable Cryptographic Cortex-MO Processor With In-Memory and Near-Memory Computing for IoT Security

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Yiqun; Xu, Li; Dong, Qing; Wang, Jingcheng; Blaauw, David; Sylvester, Dennis

被引用次数: 83

2018

A 4+2T SRAM for Searching and In-Memory Computing With 0.3-V VDDmin

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Dong, Qing; Jeloka, Supreet; Saligane, Mehdi; Kim, Yejoong; Kawaminami, Masaru; Harada, Akihiko; Miyoshi, Satoru; Yasuda, Makoto; Blaauw, David; Sylvester, Dennis

被引用次数: 71

2018

iRazor: Current-Based Error Detection and Correction Scheme for PVT Variation in 40-nm ARM Cortex-R4 Processor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Yiqun; Khayatzadeh, Mahmood; Yang, Kaiyuan; Saligane, Mehdi; Pinckney, Nathaniel; Alioto, Massimo; Blaauw, David; Sylvester, Dennis

被引用次数: 66

2018

A Noise Reconfigurable All-Digital Phase-Locked Loop Using a Switched Capacitor-Based Frequency-Locked Loop and a Noise Detector

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jang, Taekwang; Jeong, Seokhyeon; Jeon, Dongsuk; Choo, Kyojin David; Sylvester, Dennis; Blaauw, David

被引用次数: 19

2018

A 1920 x 1080 30-frames/s 2.3 TOPS/W Stereo-Depth Processor for Energy-Efficient Autonomous Navigation of Micro Aerial Vehicles

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Ziyun; Dong, Qing; Saligane, Mehdi; Kempke, Benjamin; Gong, Luyao; Zhang, Zhengya; Dreslinski, Ronald; Sylvester, Dennis; Blaauw, David; Kim, Hun-Seok

被引用次数: 17

2018

Always-On 12-nW Acoustic Sensing and Object Recognition Microsystem for Unattended Ground Sensor Nodes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Seokhyeon; Chen, Yu; Jang, Taekwang; Tsai, Julius Ming-Lin; Blaauw, David; Kim, Hun-Seok; Sylvester, Dennis

被引用次数: 59

2018

Hardware Designs for Security in Ultra-Low-Power IoT Systems: An Overview and Survey

IEEE MICRO

Yang, Kaiyuan; Blaauw, David; Sylvester, Dennis

被引用次数: 57

2017

Exploiting the Analog Properties of Digital Circuits for Malicious Hardware

COMMUNICATIONS OF THE ACM

Yang, Kaiyuan; Hicks, Matthew; Dong, Qing; Austin, Todd; Sylvester, Dennis

被引用次数: 4

2017

A Fully Integrated Counter Flow Energy Reservoir for Peak Power Delivery in Small Form-Factor Sensor Systems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wu, Xiao; Choo, Kyojin; Shi, Yao; Chuo, Li-Xuan; Sylvester, Dennis; Blaauw, David

被引用次数: 1

2017

A 23-mW Face Recognition Processor with Mostly-Read 5T Memory in 40-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeon, Dongsuk; Dong, Qing; Kim, Yejoong; Wang, Xiaolong; Chen, Shuai; Yu, Hao; Blaauw, David; Sylvester, Dennis

被引用次数: 30

2017

A Subthreshold Voltage Reference With Scalable Output Voltage for Low-Power IoT Systems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Inhee; Sylvester, Dennis; Blaauw, David

被引用次数: 163

2017

A 20-pW Discontinuous Switched-Capacitor Energy Harvester for Smart Sensor Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wu, Xiao; Shi, Yao; Jeloka, Supreet; Yang, Kaiyuan; Lee, Inhee; Lee, Yoonmyung; Sylvester, Dennis; Blaauw, David

被引用次数: 54

2017

Edge-Pursuit Comparator: An Energy-Scalable Oscillator Collapse-Based Comparator With Application in a 74.1 dB SNDR and 20 kS/s 15 b SAR ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shim, Minseob; Jeong, Seokhyeon; Myers, Paul D.; Bang, Suyoung; Shen, Junhua; Kim, Chulwoo; Sylvester, Dennis; Blaauw, David; Jung, Wanyeong

被引用次数: 53

2017

Introduction to the January Special Issue on the 2016 IEEE International Solid-State Circuits Conference

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sylvester, Dennis; Markovic, Dejan; Genov, Roman; Kawasumi, Atsushi; Mitra, Subhasish

被引用次数: 0

2017

Circuit and System Designs of Ultra-Low Power Sensor Nodes With Illustration in a Miniaturized GNSS Logger for Position Tracking: Part I-Analog Circuit Techniques

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jang, Taekwang; Kim, Gyouho; Kempke, Benjamin; Henry, Michael B.; Chiotellis, Nikolaos; Pfeiffer, Carl; Kim, Dongkwun; Kim, Yejoong; Foo, Zhiyoong; Kim, Hyeongseok; Grbic, Anthony; Sylvester, Dennis; Kim, Hun-Seok; Wentzloff, David D.; Blaauw, David

被引用次数: 27

2017

Low-Power and Compact Analog-to-Digital Converter Using Spintronic Racetrack Memory Devices

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Dong, Qing; Yang, Kaiyuan; Fick, Laura; Fick, David; Blaauw, David; Sylvester, Dennis

被引用次数: 16

2017

Impact of FinFET on Near-Threshold Voltage Scalability

IEEE DESIGN & TEST

Pinckney, Nathaniel; Jeloka, Supreet; Dreslinski, Ron; Mudge, Trevor; Sylvester, Dennis; Blaauw, David; Shifren, Lucian; Cline, Brian; Sinha, Saurabh

被引用次数: 11

2017

A Wire-overhead-free Reset Propagation Scheme for Millimeter-scale Sensor Systems

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Inhee; Bang, Suyoung; Kim, Yejoong; Kim, Gyouho; Sylvester, Dennis; Blaauw, David; Lee, Yoonmyung

被引用次数: 2

2017

Ultralow Power Circuit Design for Wireless Sensor Nodes for Structural Health Monitoring

PROCEEDINGS OF THE IEEE

Lee, Yoonmyung; Blaauw, David; Sylvester, Dennis

被引用次数: 53

2016

A Resonant Current-Mode Wireless Power Receiver and Battery Charger With-32 dBm Sensitivity for Implantable Systems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Myungjoon; Jang, Taekwang; Jeong, Junwon; Jeong, Seokhyeon; Blaauw, David; Sylvester, Dennis

被引用次数: 45

2016

Battery Voltage Supervisors for Miniature IoT Systems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Inhee; Lee, Yoonmyung; Sylvester, Dennis; Blaauw, David

被引用次数: 25

2016

A 110 nW Resistive Frequency Locked On-Chip Oscillator with 34.3 ppm/°C Temperature Stability for System-on-Chip Designs

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Myungjoon; Jang, Taekwang; Bang, Suyoung; Shi, Yao; Blaauw, David; Sylvester, Dennis

被引用次数: 61

2016

A Low Ripple Switched-Capacitor Voltage Regulator Using Flying Capacitance Dithering

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bang, Suyoung; Seo, Jae-sun; Chang, Leland; Blaauw, David; Sylvester, Dennis

被引用次数: 25

2016

A 28 nm Configurable Memory (TCAM/BCAM/SRAM) Using Push-Rule 6T Bit Cell Enabling Logic-in-Memory

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeloka, Supreet; Akesh, Naveen Bharathwaj; Sylvester, Dennis; Blaauw, David

被引用次数: 270

2016

An All-Digital Edge Racing True Random Number Generator Robust Against PVT Variations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Kaiyuan; Blaauw, David; Sylvester, Dennis

被引用次数: 108

2016

A Constant Energy-Per-Cycle Ring Oscillator Over a Wide Frequency Range for Wireless Sensor Nodes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Inhee; Sylvester, Dennis; Blaauw, David

被引用次数: 53

2016

A 5.58 nW Crystal Oscillator Using Pulsed Driver for Real-Time Clocks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yoon, Dongmin; Jang, Taekwang; Sylvester, Dennis; Blaauw, David

被引用次数: 36

2016

A Successive-Approximation Switched-Capacitor DC-DC Converter With Resolution of $V_{IN}/2N$ for a Wide Range of Input and Output Voltages

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bang, Suyoung; Blaauw, David; Sylvester, Dennis

被引用次数: 62

2016

Approximate SRAMs With Dynamic Energy-Quality Management

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Frustaci, Fabio; Blaauw, David; Sylvester, Dennis; Alioto, Massimo

被引用次数: 33

2016

A 5.8 nW CMOS Wake-Up Timer for Ultra-Low-Power Wireless Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Seokhyeon; Lee, Inhee; Blaauw, David; Sylvester, Dennis

被引用次数: 53

2015

A Dual-Slope Capacitance-to-Digital Converter Integrated in an Implantable Pressure-Sensing System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Sechang; Lee, Yoonmyung; Wang, Jingcheng; Foo, Zhiyong; Kim, Yejoong; Jung, Wanyoung; Li, Ziyun; Blaauw, David; Sylvester, Dennis

被引用次数: 74

2015

Digitally Controlled Leakage-Based Oscillator and Fast Relocking MDLL for Ultra Low Power Sensor Platform

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jee, Dong-Woo; Sylvester, Dennis; Blaauw, David; Sim, Jae-Yoon

被引用次数: 9

2015

SRAM for Error-Tolerant Applications With Dynamic Energy-Quality Management in 28 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Frustaci, Fabio; Khayatzaadeh, Mahmood; Blaauw, David; Sylvester, Dennis; Alioto, Massimo

被引用次数: 59

2015

An Injectable 64 nW ECG Mixed-Signal SoC in 65 nm for Arrhythmia Monitoring

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Yen-Po; Jeon, Dongsuk; Lee, Yoonmyung; Kim, Yejoong; Foo, Zhiyoong; Lee, Inhee; Langhals, Nicholas B.; Kruger, Grant; Oral, Hakan; Berenfeld, Omer; Zhang, Zhengya; Blaauw, David; Sylvester, Dennis

被引用次数: 142

2015

System-On-Mud: Ultra-Low Power Oceanic Sensing Platform Powered by Small-Scale Benthic Microbial Fuel Cells

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Inhee; Kim, Gyouho; Bang, Suyoung; Wolfe, Adriane; Bell, Richard; Jeong, Seokhyeon; Kim, Yejoong; Kagan, Jeffrey; Arias-Thode, Meriah; Chadwick, Bart; Sylvester, Dennis; Blaauw, David; Lee, Yoonmyung

被引用次数: 33

2015

An Ultra-Low Power Fully Integrated Energy Harvester Based on Self-Oscillating Switched-Capacitor Voltage Doubler

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jung, Wanyoung; Oh, Sechang; Bang, Suyoung; Lee, Yoonmyung; Foo, Zhiyoong; Kim, Gyouho; Zhang, Yiqun; Sylvester, Dennis; Blaauw, David

被引用次数: 134

2014

A 346 μm^2 VCO-Based, Reference-Free, Self-Timed Sensor Interface for Cubic-Millimeter Sensor Nodes in 28 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fick, Laura; Fick, David; Alioto, Massimo; Blaauw, David; Sylvester, Dennis

被引用次数: 23

2014

Razor-Lite: A Light-Weight Register for Error Detection by Observing Virtual Supply Rails

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kwon, Inyong; Kim, Seongjong; Fick, David; Kim, Myungbo; Chen, Yen-Po; Sylvester, Dennis

被引用次数: 70

2014

A Fully-Integrated 71 nW CMOS Temperature Sensor for Low Power Wireless Sensor Nodes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Seokhyeon; Foo, Zhiyoong; Lee, Yoonmyung; Sim, Jae-Yoon; Blaauw, David; Sylvester, Dennis

被引用次数: 177

2014

An Energy Efficient Full-Frame Feature Extraction Accelerator With Shift-Latch FIFO in 28 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeon, Dongsuk; Henry, Michael B.; Kim, Yejoong; Lee, Inhee; Zhang, Zhengya; Blaauw, David; Sylvester, Dennis

被引用次数: 28

2014

Low-Power High-Throughput LDPC Decoder Using Non-Refresh Embedded DRAM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Youn Sung; Blaauw, David; Sylvester, Dennis; Zhang, Zhengya

被引用次数: 64

2014

Design and Evaluation of Confidence-Driven Error-Resilient Systems

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chen, Chia-Hsiang; Blaauw, David; Sylvester, Dennis; Zhang, Zhengya

被引用次数: 6

2014

A Statistical Framework for Post-Fabrication Oxide Breakdown Reliability Prediction and Management

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhuo, Cheng; Sylvester, Dennis; Blaauw, David

被引用次数: 3

2013

Underdesigned and Opportunistic Computing in Presence of Hardware Variability

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Gupta, Puneet; Agarwal, Yuvraj; Dolecek, Lara; Dutt, Nikil; Gupta, Rajesh K.; Kumar, Rakesh; Mitra, Subhasish; Nicolau, Alexandru; Rosing, Tajana Simunic; Srivastava, Mani B.; Swanson, Steven; Sylvester, Dennis

被引用次数: 107

2013

LIMITS OF PARALLELISM AND BOOSTING IN DIM SILICON

IEEE MICRO

Pinckney, Nathaniel; Dreslinski, Ronald G.; Sewell, Korey; Fick, David; Mudge, Trevor; Sylvester, Dennis; Blaauw, David

被引用次数: 9

2013

CENTIP3DE: A 64-CORE, 3D STACKED NEAR-THRESHOLD SYSTEM

IEEE MICRO

Dreslinski, Ronald G.; Fick, David; Giridhar, Bharan; Kim, Gyouho; Seo, Sangwon; Fojtik, Matthew; Satpathy, Sudhir; Lee, Yoonmyung; Kim, Daeyeon; Liu, Nurrachman; Wieckowski, Michael; Chen, Gregory; Sylvester, Dennis; Blaauw, David; Mudge, Trevor

被引用次数: 32

2013

Centip3De: A Many-Core Prototype Exploring 3D Integration and Near-Threshold Computing

COMMUNICATIONS OF THE ACM

Dreslinski, Ronald G.; Fick, David; Giridhar, Bharan; Kim, Gyouho; Seo, Sangwon; Fojtik, Matthew; Satpathy, Sudhir; Lee, Yoonmyung; Kim, Daeyeon; Liu, Nurrachman; Wieckowski, Michael; Chen, Gregory; Sylvester, Dennis; Blaauw, David; Mudge, Trevor

被引用次数: 7

2013

A Sub-nW Multi-stage Temperature Compensated Timer for Ultra-Low-Power Sensor Nodes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Yoonmyung; Giridhar, Bharan; Foo, Zhiyoong; Sylvester, Dennis; Blaauw, David B.

被引用次数: 49

2013

A Millimeter-Scale Energy-Autonomous Sensor System With Stacked Battery and Solar Cells

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fojtik, Matthew; Kim, Daeyeon; Chen, Gregory; Lin, Yu-Shiang; Fick, David; Park, Junsun; Seok, Mingoo; Chen, Mao-Ter; Foo, Zhiyoong; Blaauw, David; Sylvester, Dennis

被引用次数: 94

2013

Bubble Razor: Eliminating Timing Margins in an ARM Cortex-M3 Processor in 45 nm CMOS Using Architecturally Independent Error Detection and Correction

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fojtik, Matthew; Fick, David; Kim, Yejoong; Pinckney, Nathaniel; Harris, David Money; Blaauw, David; Sylvester, Dennis

被引用次数: 141

2013

Centip3De: A Cluster-Based NTC Architecture With 64 ARM Cortex-M3 Cores in 3D Stacked 130 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fick, David; Dreslinski, Ronald G.; Giridhar, Bharan; Kim, Gyouho; Seo, Sangwon; Fojtik, Matthew; Satpathy, Sudhir; Lee, Yoonmyung; Kim, Daeyeon; Liu, Nurrachman; Wieckowski, Michael; Chen, Gregory; Mudge, Trevor; Blaauw, David; Sylvester, Dennis

被引用次数: 38

2013

A Modular 1 mm³ Die-Stacked Sensing Platform With Low Power I2C Inter-Die Communication and Multi-Modal Energy Harvesting

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Yoonmyung; Bang, Suyoung; Lee, Inhee; Kim, Yejoong; Kim, Gyouho; Ghaed, Mohammad Hassan; Pannuto, Pat; Dutta, Prabal; Sylvester, Dennis; Blaauw, David

被引用次数: 181

2013

Circuits for a Cubic-Millimeter Energy-Autonomous Wireless Intraocular Pressure Monitor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ghaed, Mohammad Hassan; Chen, Gregory; Razi-ul Haque; Wieckowski, Michael; Kim, Yejoong; Kim, Gyouho; Lee, Yoonmyung; Lee, Inhee; Fick, David; Kim, Daeyeon; Seok, Mingoo; Wise, Kensall D.; Blaauw, David; Sylvester, Dennis

被引用次数: 102

2013

Achieving Ultralow Standby Power With an Efficient SCCMOS Bias Generator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Yoonmyung; Seok, Mingoo; Hanson, Scott; Sylvester, Dennis; Blaauw, David

被引用次数: 5

2013

Introduction to the Special Section on Circuits and Systems for Energy-Autonomous Microsystems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sylvester, Dennis

被引用次数: 0

2013

Low-Power Circuit Analysis and Design Based on Heterojunction Tunneling Transistors (HETTs)

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, Yoonmyung; Kim, Daeyeon; Cai, Jin; Lauer, Isaac; Chang, Leland; Koester, Steven J.; Blaauw, David; Sylvester, Dennis

被引用次数: 54

2013

A Reliable Routing Architecture and Algorithm for NoCs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

DeOrio, Andrew; Fick, David; Bertacco, Valeria; Sylvester, Dennis; Blaauw, David; Hu, Jin; Chen, Gregory

被引用次数: 80

2012

A Portable 2-Transistor Picowatt Temperature-Compensated Voltage Reference Operating at 0.5 V

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Seok, Mingoo; Kim, Gyouho; Blaauw, David; Sylvester, Dennis

被引用次数: 349

2012

A Super-Pipelined Energy Efficient Subthreshold 240 MS/s FFT Core in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeon, Dongsuk; Seok, Mingoo; Chakrabarti, Chaitali; Blaauw, David; Sylvester, Dennis

被引用次数: 66

2012

Design Methodology for Voltage-Overscaled Ultra-Low-Power Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jeon, Dongsuk; Seok, Mingoo; Zhang, Zhengya; Blaauw, David; Sylvester, Dennis

被引用次数: 22

2012

Swizzle-Switch Networks for Many-Core Systems

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Sewell, Korey; Dreslinski, Ronald G.; Manville, Thomas; Satpathy, Sudhir; Pinckney, Nathaniel; Blake, Geoffrey; Cieslak, Michael; Das, Reetuparna; Wenisch, Thomas F.; Sylvester, Dennis; Blaauw, David; Mudge, Trevor

被引用次数: 65

2012

Compact Degradation Sensors for Monitoring NBTI and Oxide Degradation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Singh, Prashant; Karl, Eric; Blaauw, David; Sylvester, Dennis

被引用次数: 37

2012

Sleep Mode Analysis and Optimization With Minimal-Sized Power Gating Switch for Ultra-Low Vdd Operation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Seok, Mingoo; Hanson, Scott; Blaauw, David; Sylvester, Dennis

被引用次数: 10

2012

Process Variation and Temperature-Aware Full Chip Oxide Breakdown Reliability Analysis

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhuo, Cheng; Chopra, Kaviraj; Sylvester, Dennis; Blaauw, David

被引用次数: 18

2011

Fast Statistical Static Timing Analysis Using Smart Monte Carlo Techniques

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Veetil, Vineeth; Chopra, Kaviraj; Blaauw, David; Sylvester, Dennis

被引用次数: 24

2011

Crosstalk-Aware PWM-Based On-Chip Links With Self-Calibration in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Seo, Jae-sun; Blaauw, David; Sylvester, Dennis

被引用次数: 9

2011

Dynamic NBTI Management Using a 45 nm Multi-Degradation Sensor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Singh, Prashant; Karl, Eric; Sylvester, Dennis; Blaauw, David

被引用次数: 29

2011

Robust Clock Network Design Methodology for Ultra-Low Voltage Operations

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Seok, Mingoo; Blaauw, David; Sylvester, Dennis

被引用次数: 15

2011

CAS-FEST 2010: Mitigating Variability in Near-Threshold Computing

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Seok, Mingoo; Chen, Gregory; Hanson, Scott; Wieckowski, Michael; Blaauw, David; Sylvester, Dennis

被引用次数: 44

2011

A Robust Edge Encoding Technique for Energy-Efficient Multi-Cycle Interconnect

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Seo, Jae-Sun; Kaul, Himanshu; Krishnamurthy, Ram; Sylvester, Dennis; Blaauw, David

被引用次数: 9

2011

Circuit Design Advances for Wireless Sensing Applications

PROCEEDINGS OF THE IEEE

Chen, Gregory; Hanson, Scott; Blaauw, David; Sylvester, Dennis

被引用次数: 51

2010

Near-Threshold Computing: Reclaiming Moore's Law Through Energy Efficient Integrated Circuits

PROCEEDINGS OF THE IEEE

Dreslinski, Ronald G.; Wieckowski, Michael; Blaauw, David; Sylvester, Dennis; Mudge, Trevor

被引用次数: 644

2010

Mechanical Stress Aware Optimization for Leakage Power Reduction

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Joshi, Vivek; Cline, Brian; Sylvester, Dennis; Blaauw, David; Agarwal, Kanak

被引用次数: 10

2010

Victim Alignment in Crosstalk-Aware Timing Analysis

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Gandikota, Ravikishore; Chopra, Kaviraj; Blaauw, David; Sylvester, Dennis

被引用次数: 5

2010

A 0.5 V Sub-Microwatt CMOS Image Sensor With Pulse-Width Modulation Read-Out

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hanson, Scott; Foo, ZhiYoong; Blaauw, David; Sylvester, Dennis

被引用次数: 76

2010

Yield-Driven Near-Threshold SRAM Design

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chen, Gregory; Sylvester, Dennis; Blaauw, David; Mudge, Trevor

被引用次数: 72

2010

Dynamically Pulsed MTCMOS With Bus Encoding for Reduction of Total Power and Crosstalk Noise

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Singh, Harmander; Rao, Rahul; Agarwal, Kanak; Sylvester, Dennis; Brown, Richard

被引用次数: 5

2010

Circuit Optimization Techniques to Mitigate the Effects of Soft Errors in Combinational Logic

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Rao, Rajeev R.; Joshi, Vivek; Blaauw, David; Sylvester, Dennis

被引用次数: 9

2009

A Low-Voltage Processor for Sensing Applications With Picowatt Standby Mode

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hanson, Scott; Seok, Mingoo; Lin, Yu-Shiang; Foo, ZhiYoong; Kim, Daeyeon; Lee, Yoonmyung; Liu, Nurrachman; Sylvester, Dennis; Blaauw, David

被引用次数: 132

2009

Alignment-Independent Chip-to-Chip Communication for Sensor Applications Using Passive Capacitive Signaling

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Yu-Shiang; Sylvester, Dennis; Blaauw, David

被引用次数: 11

2009

Energy-Efficient Subthreshold Processor Design

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhai, Bo; Pant, Sanjay; Nazhandali, Leyla; Hanson, Scott; Olson, Javin; Reeves, Anna; Minuth, Michael; Helfand, Ryan; Austin, Todd; Sylvester, Dennis; Blaauw, David

被引用次数: 133

2009

Simple and Accurate Models for Capacitance Considering Floating Metal Fill Insertion

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Youngmin; Petranovic, Dusan; Sylvester, Dennis

被引用次数: 9

2009

A novel approach to perform gate-level yield analysis and optimization considering correlated variations in power and performance

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Srivastava, Ashish; Chopra, Kaviraj; Shah, Saumil; Sylvester, Dennis; Blaauw, David

被引用次数: 19

2008

A Variation-Tolerant Sub-200 mV 6-T Subthreshold SRAM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhai, Bo; Hanson, Scott; Blaauw, David; Sylvester, Dennis

被引用次数: 132

2008

Exploring variability and performance in a sub-200-mV processor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hanson, Scott; Zhai, Bo; Seok, Mingoo; Cline, Brian; Zhou, Kevin; Singhal, Meghna; Minuth, Michael; Olson, Javin; Nazhandali, Leyla; Austin, Todd; Sylvester, Dennis; Blaauw, David

被引用次数: 119

2008

Self-timed regenerators for high-speed and low-power on-chip global interconnect

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Singh, Prashant; Seo, Jae-Sun; Blaauw, David; Sylvester, Dennis

被引用次数: 10

2008

Multi-mechanism reliability modeling and management in dynamic systems

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Karl, Eric; Blaauw, David; Sylvester, Dennis; Mudge, Trevor

被引用次数: 42

2008

Nanometer device scaling in subthreshold logic and SRAM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hanson, Scott; Seok, Mingoo; Sylvester, Dennis; Blaauw, David

被引用次数: 112

2008

Variability in nanometer CMOS: Impact, analysis, and minimization

INTEGRATION-THE VLSI JOURNAL

Sylvester, Dennis; Agarwal, Kanak; Shah, Saumil

被引用次数: 49

2008

Compact HSPICE model for IMOS device

ELECTRONICS LETTERS

Lin, J.; Toh, E. H.; Shen, C.; Sylvester, D.; Heng, C. H.; Samudra, G.; Yeo, Y. C.

被引用次数: 10

2008

Computer-aided design for low-power robust computing in nanoscale CMOS

PROCEEDINGS OF THE IEEE

Sylvester, Dennis; Srivastava, Ashish

被引用次数: 14

2007

Self-compensating design for reduction of timing and leakage sensitivity to systematic pattern-dependent variation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Gupta, Puneet; Kahng, Andrew B.; Kim, Youngmin; Sylvester, Dennis

被引用次数: 5

2007

Enhanced leakage reduction techniques using intermediate strength power gating

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Singh, Harmander; Agarwal, Kanak; Sylvester, Dennis; Nowka, Kevin J.

被引用次数: 89

2007

Parametric yield analysis and optimization in leakage dominated technologies

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Agarwal, Kanak; Rao, Rahul; Sylvester, Dennis; Brown, Richard

被引用次数: 16

2007

Analytical yield prediction considering leakage/performance correlation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Rao, Rajeev R.; Devgan, Anirudh; Blaauw, David; Sylvester, Dennis

被引用次数: 22

2006

Gate-length biasing for runtime-leakage control

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Gupta, Puneet; Kahng, Andrew B.; Sharma, Puneet; Sylvester, Dennis

被引用次数: 86

2006

Statistical interconnect metrics for physical-design optimization

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Agarwal, Kanak; Agarwal, Mridul; Sylvester, Dennis; Blaauw, David

被引用次数: 111	2006
Runtime leakage minimization through probability-aware optimization	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Lee, Dongwoo; Blaauw, David; Sylvester, Dennis	
被引用次数: 4	2006
Modeling and analysis of crosstalk noise in coupled RLC interconnects	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Agarwal, K; Sylvester, D; Blaauw, D	
被引用次数: 72	2005
Bus encoding for total power reduction using a leakage-aware buffer configuration	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Rao, RR; Deogan, HS; Blaauw, D; Sylvester, D	
被引用次数: 14	2005
Design and analysis of spatial encoding circuits for peak power reduction in on-chip buses	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Kaul, H; Sylvester, D; Anders, MA; Krishnamurthy, RK	
被引用次数: 16	2005
The limit of dynamic voltage scaling and insomniac dynamic voltage scaling	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Zhai, B; Blaauw, D; Sylvester, D; Flautner, K	
被引用次数: 87	2005
Switch-factor based loop RLC modeling for efficient timing analysis	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Cao, Y; Yang, XD; Huang, XJ; Sylvester, D	
被引用次数: 9	2005
Minimizing total power by simultaneous Vdd/Vth assignment	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Srivastava, A; Sylvester, D	
被引用次数: 23	2004
A library compatible driver output model for on-chip RLC transmission lines	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Agarwal, K; Sylvester, D; Blaauw, D	
被引用次数: 5	2004
Performance optimization of critical nets through active shielding	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Kaul, H; Sylvester, D; Blaauw, D	
被引用次数: 28	2004

Low-power on-chip communication based on transition-aware global signaling (TAGS)

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kaul, H; Sylvester, D

被引用次数: 21

2004

Statistical analysis of subthreshold leakage current for VLSI circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Rao, R; Srivastava, A; Blaauw, D; Sylvester, D

被引用次数: 123

2004

Gate oxide leakage current analysis and reduction for VLSI circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, D; Blaauw, D; Sylvester, D

被引用次数: 98

2004

Bidirectional closed-form transformation between on-chip coupling noise waveforms and interconnect delay-change curves

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Sato, T; Cao, Y; Agarwal, K; Sylvester, D; Hu, CM

被引用次数: 23

2003

Improved a priori interconnect predictions and technology extrapolation in the GTX system

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Cao, Y; Hu, CM; Huang, XJ; Kahng, AB; Markov, IL; Oliver, M; Stroobandt, D; Sylvester, D

被引用次数: 16

2003

Effective on-chip inductance modeling for multiple signal lines and application to repeater insertion

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Cao, Y; Huang, XJ; Chang, NH; Lin, S; Nakagawa, OS; Xie, WZ; Sylvester, D; Hu, CM

被引用次数: 24

2002

Analytical modeling and characterization of deep-submicrometer interconnect

PROCEEDINGS OF THE IEEE

Sylvester, D; Hu, CM

被引用次数: 131

2001

Impact of small process geometries on microarchitectures in systems on a chip

PROCEEDINGS OF THE IEEE

Sylvester, D; Keutzer, K

被引用次数: 86

2001

Accurate in situ measurement of peak noise and delay change induced by interconnect coupling

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sato, T; Sylvester, D; Cao, Y; Hu, CM

被引用次数: 25

2001

A global wiring paradigm for deep submicron design

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Sylvester, D; Keutzer, K

被引用次数: 111

2000

Rethinking deep-submicron circuit design

COMPUTER

Sylvester, D; Keutzer, K

被引用次数: 28

1999

Investigation of interconnect capacitance characterization using charge-based capacitance measurement (CBCM) technique and three-dimensional simulation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sylvester, D; Chen, JC; Hu, CM

被引用次数: 59

1998

An on-chip, interconnect capacitance characterization method with sub-femto-farad resolution

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Chen, JC; Sylvester, D; Hu, CM

被引用次数: 40

1998

Cache Automaton

MICRO

Arun Subramaniyan; Jingcheng Wang; R. M. Ezhil Balasubramanian; David Blaauw; Dennis Sylvester; Reetuparna Das

被引用次数: 0

Ultralow Power Circuit Design for Wireless Sensor Nodes for Structural Health Monitoring

Proc.IEEE

Yoonmyung Lee; David Blaauw; Dennis Sylvester

被引用次数: 0

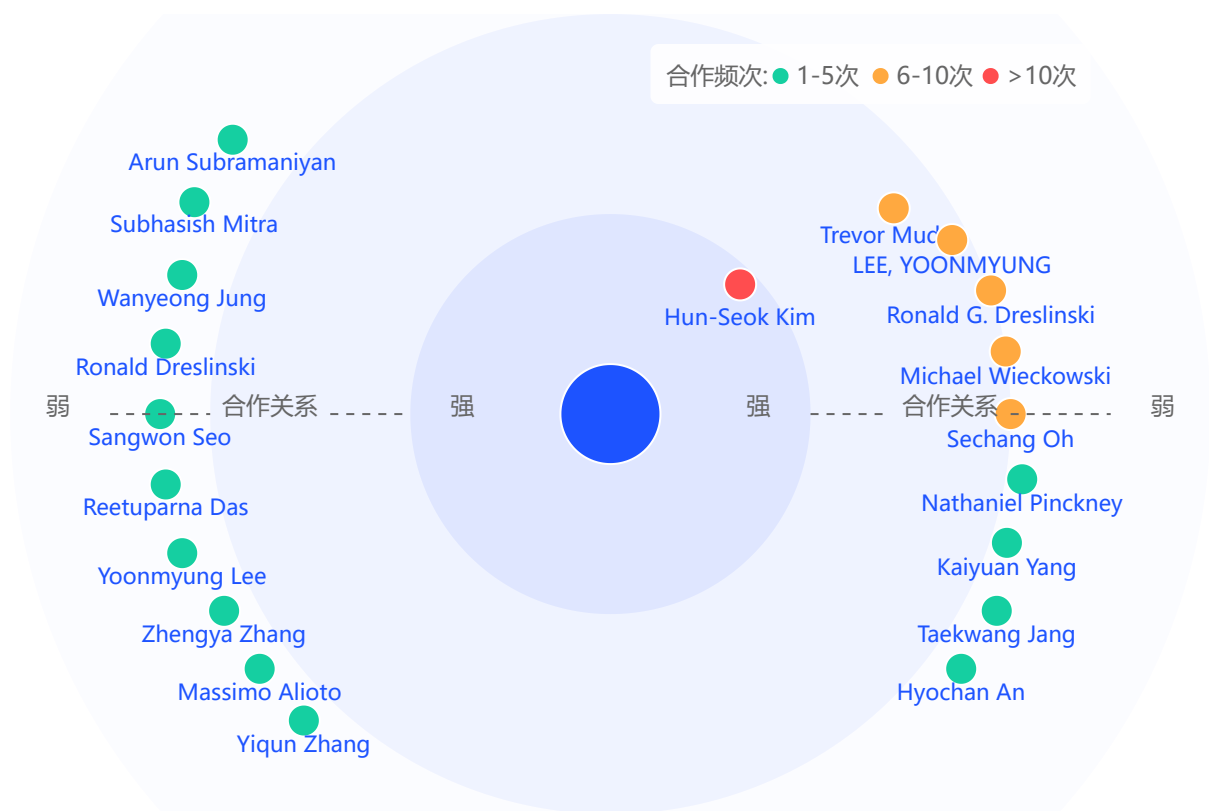
A2: Analog Malicious Hardware

S&P

Kaiyuan Yang; Matthew Hicks; Qing Dong; Todd Austin; Dennis Sylvester

被引用次数: 0

学术合作



Hun-Seok Kim

合作次数: 19

密歇根大学-安娜堡

Trevor Mudge

合作次数: 9

密歇根大学-安娜堡

LEE, YOONMYUNG

合作次数: 7

成均馆大学

Ronald G. Dreslinski

合作次数: 6

密歇根大学-安娜堡

Michael Wieckowski

合作次数: 6

密歇根大学-安娜堡

还有 48 个合作者